

MAT 2305- Real Analysis II
Assignment 04

Answers all the questions and submit the **solutions** for **1, 2[i, iii, v, iX]** on or before **2nd August , 2024**

1. Evaluate each of the following integrals:

- i. $\iint_R (2x + 3y) dA$, where R is the rectangle region defined by $0 \leq x \leq 2$ and $1 \leq y \leq 3$.
- ii. $\iint_R (x^2 + y^2) dA$, where R is the region bounded by $y = x^2$ and $y = 2 - x^2$.
- iii. $\iint_R e^{x^2+y^2} dA$, where R is the disk of radius 2 and centered at the origin.
- iv. $\iint_R \frac{1}{1+x^2+y^2} dA$, R is the region inside the unit disk given by $x^2 + y^2 \leq 1$.
- v. $\iint_R \sin(x^2 + y^2) dA$, R is the annular region between the circle of radius 1 and radius at the origin.
- vi. $\iint_R x e^{y^2} dA$, where R is the triangular region with vertices: $(0, 0), (1, 0), (1, 1)$.

2. Evaluate each of the following integrals:

- i. $\iiint_D (x + y + z) dV$, D is the cuboid defined by $0 \leq x \leq 1, 0 \leq y \leq 1$, and $0 \leq z \leq 1$.
- ii. $\iiint_D z dV$, where D is the region bounded by the planes: $z = 0, z = 1 - x, y = 0$, and $y = 1 - x$.
- iii. $\iiint_D (x^2 + y^2) dV$, where D is the cylinder defined by $x^2 + y^2 \leq 4$ and $0 \leq z \leq 3$.
- iv. $\iiint_D \rho^2 \sin \phi dV$, where D is the sphere of radius 3 centered at the origin.
- v. $\iiint_D z \sin(x^2 + y^2) dV$, where D is the solid region defined by $x^2 + y^2 \leq 4$ and $0 \leq z \leq \sqrt{4 - x^2 - y^2}$.
- vi. $\iiint_D r^3 \cos \theta dV$, where D is the cone defined by $z^2 = x^2 + y^2$ and $0 \leq z \leq 3$.
- vii. $\iiint_D \frac{1}{\rho^2 \sin \phi} dV$, where D is the sphere shell defined by $1 \leq \rho \leq 2$.
- viii. $\iiint_D \sin(\rho \cos \phi) dV$, where D is the upper hemisphere of radius 3 centered at the origin.
- ix. $\iiint_D \sqrt{x^2 + y^2 + z^2} dV$, D is the region inside the sphere but outside the cylinder $x^2 + y^2 = 1$.
- x. $\iiint_D z dV$, where D is the tetrahedron bounded by the coordinate planes and the plane $x + y + z = 1$.